

IT Infrastructure and Emerging Technologies

From the chip inside your laptop to the cloud

Chapter 5 · Laudon & Laudon, Management Information Systems (18th ed.)

Today's roadmap

01	**What is a computer?**	Hardware basics: CPU, memory, storage, operating systems
02	**CPU vs GPU**	Two kinds of processors — and why AI needs GPUs
03	**IT infrastructure**	What it is, how it evolved, its seven components
04	**Mobile, virtualization & cloud**	Computing delivered as a service
05	**Trends & smart decisions**	Quantum, green IT, software choices, total cost

Opening case: QRyde rides high with the cloud and AI

Who

A cloud platform that books shared rides for people with physical or mental challenges.

Problem

Its own servers (on-premise and colocation) could no longer keep up with growth.

Solution

Moved to Oracle Cloud Infrastructure, added AI route planning, pays only for what it uses.

Think about it: why would a growing company stop owning its own computers?

100,000

trips coordinated per day

700+

organizations using the platform

Pay per use

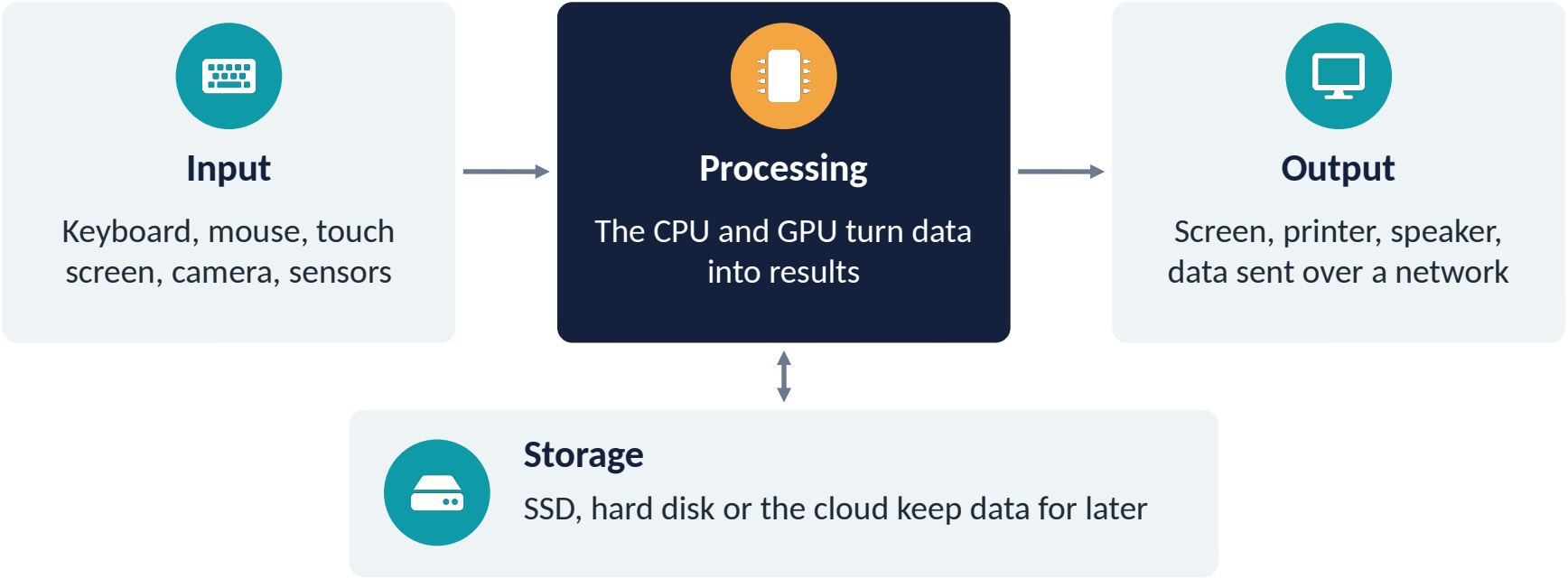
instead of large up-front IT investments

01

What is a computer, really?

Hardware basics for absolute beginners

Every computer does four things



Example: you type prices (input) → the spreadsheet adds VAT (processing) → the total appears (output) → you save the file (storage).

Hardware vs software

Hardware

- Physical parts you can touch
- Chips, screens, keyboards, cables, servers
- Wears out and must be replaced

Software

- Instructions that tell the hardware what to do
- Operating systems, apps, websites
- Can be copied, updated or rented online



Analogy: hardware is the piano, software is the sheet music — you need both to hear a song.

What is inside a computer?

CPU (processor)

The "brain": follows program instructions step by step

RAM (memory)

Short-term working memory: fast, but erased when power is off

Storage (SSD/HDD)

Long-term memory: keeps files, photos and programs

GPU (graphics)

Thousands of small cores doing many calculations at once

Motherboard

The main circuit board that connects all parts together

Network card

Connects the computer to other computers and the Internet

An easy way to remember: the computer as a kitchen

The chef → CPU	Reads the recipe and does the work, one step after another
The countertop → RAM	Space for what you're working on now — bigger = more at once
The pantry → Storage	Keeps ingredients for later: slower to reach, but holds a lot
1,000 helpers → GPU	Each chops one onion — together they finish in seconds
The recipe → Software	Tells the kitchen exactly what to do and in what order

Bits and bytes: how computers count



A bit is a single 0 or 1 — like a light switch. **8 bits = 1 byte = one character. 01000001 = the letter "A".**

Unit	Roughly	Everyday example
Kilobyte (KB)	1,000 bytes	A short text-only email
Megabyte (MB)	1 million bytes	One smartphone photo (about 3 MB)
Gigabyte (GB)	1 billion bytes	An HD movie (about 2–4 GB)
Terabyte (TB)	1 trillion bytes	About 250,000 photos
Petabyte (PB)	1,000 TB	Data held by a large company's data center

Memory vs storage: not the same thing

RAM (memory)

- Very fast — the CPU reads it directly
- Temporary: emptied when power is off
- Typical laptop: 8–32 GB
- More expensive per GB

VS

Storage (SSD / HDD)

- Slower (SSDs are much faster than old hard disks)
- Permanent: keeps files when power is off
- Typical laptop: 256 GB – 2 TB
- Cheaper per GB; the cloud adds even more

Tip: your laptop slows down with 40 browser tabs open? It has usually run out of RAM, not storage.

The operating system: the computer's manager

Operating system (OS): the software that manages the hardware and lets every other program run.



Desktop & laptop

- Windows — 68%+ of desktops and tablets
- macOS (Apple)
- Linux (Open Source), the best in the world :)



Smartphones & tablets

- Android — most popular worldwide
- iOS — iPhone and iPad
- Touch & multitouch interfaces



Servers & cloud

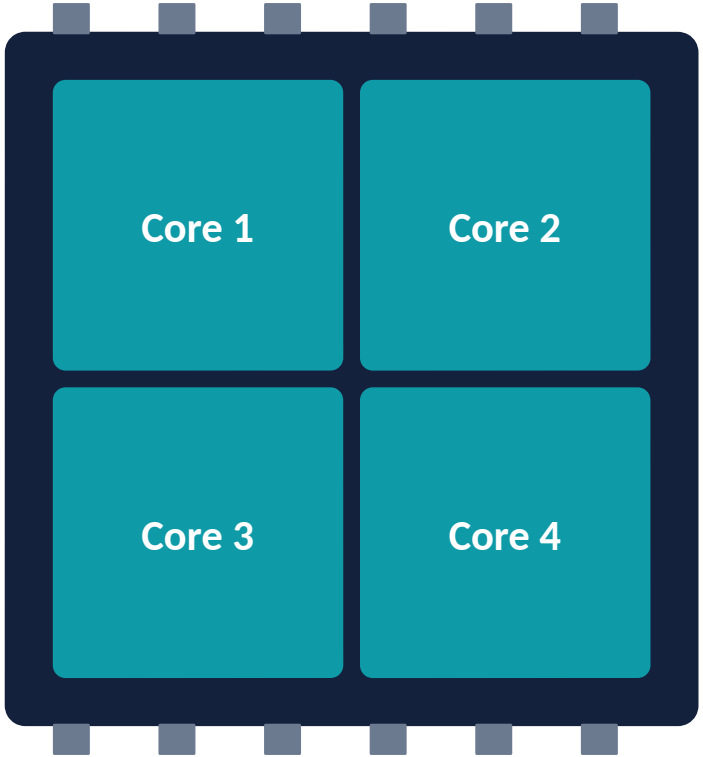
- Linux — runs about 90% of the cloud
- Windows Server
- Linux is open source: free to use

02

CPU vs GPU

Two very different kinds of "brains"

The CPU: a few very smart workers

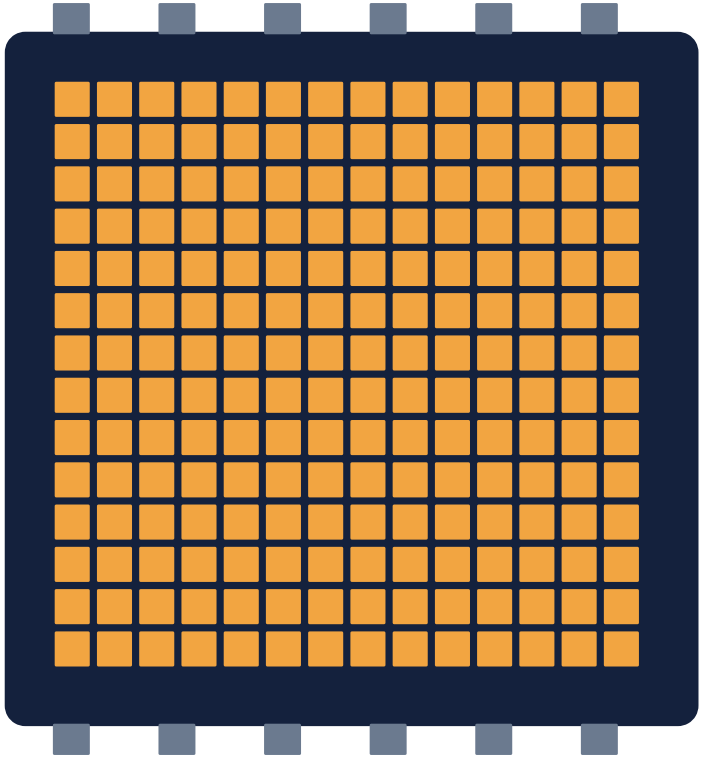


Central Processing Unit

- A few, very powerful cores (about 4–24 in a typical PC)
- Works mostly in sequence: one task after another, extremely fast
- A generalist: runs the operating system, apps, logic and decisions
- Handles all the main functions of the computer

Analogy: a small team of expert professors

The GPU: an army of simple workers



Graphics Processing Unit

- Hundreds to thousands of simpler cores
- Works in parallel: many calculations at the same moment
- Invented for graphics (Nvidia, 1999): every pixel at once
- Perfect for the same simple sum on huge amounts of data

Analogy: a stadium full of students, each solving one simple sum

CPU vs GPU side by side

	CPU	GPU
Number of cores	Few (about 4–24), very powerful	Hundreds to thousands, simpler
How it works	Serial: one task after another	Parallel: many tasks at once
Best at	Varied, complex tasks and decisions	Same simple calculation on huge data
Typical uses	OS, office apps, browsing, databases	Games, video, 3D, training AI
Analogy	A few expert professors	A stadium of students

Why the AI revolution runs on GPUs

Training an AI model means billions of simple multiplications — exactly the parallel work GPUs do best.

~80%

Nvidia's share of the market for specialized AI chips

10,000

Nvidia GPUs clustered to train ChatGPT

208 bn

transistors in Nvidia's Blackwell AI chip (2024)

Other AI chips

Google's TPUs; custom chips from Amazon, Microsoft and Apple

Chips and geopolitics

U.S. CHIPS Act (2022): $39 bn to build chip factories in the U.S.

03

IT infrastructure

The foundation every digital firm stands on

What is IT infrastructure?

"The shared technology resources that provide the platform for the firm's specific information system applications."



Technology

Hardware, software, networks and data shared across the firm



Services

People and skills: IT management, standards, training, R&D, facilities

$5.06 trillion

Worldwide IT infrastructure industry in 2024 (Gartner)

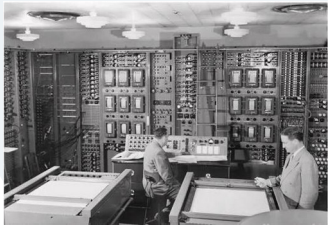
25–50%

of IT spending in large firms goes to infrastructure

Five eras of IT infrastructure

1959

Mainframe & minicomputer
Centralized giant computers, mostly IBM



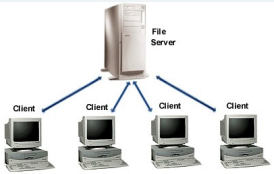
1981

Personal computer
Standalone desktops with Windows + Intel



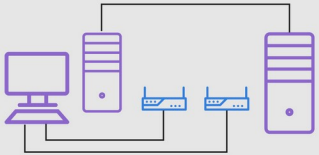
1983

Client/server
PCs networked to powerful servers



1992

Enterprise computing
Internet standards (TCP/IP) link the whole firm



2000

Cloud & mobile

Computing delivered as a service over the Internet

Client/server: how most computing works today



Client	Internet	Web server	App server	Data
Your browser or phone asks for something	Carries the request	Finds and sends the web pages	Runs business logic: prices, orders	Databases: sales, HR, accounting

Example: you search for shoes on a shop's website → the web server shows the page → the application server checks price and stock in the database → the answer comes back to your phone.

The seven components of IT infrastructure

Hardware

The physical machines

IBM, Dell, HP, Apple, NVIDIA

Operating systems

Software that runs the machines

Windows, Linux, macOS, Android, iOS

Enterprise software

Firm-wide business applications

SAP, Oracle, Microsoft

Data management

Organizes and stores the firm's data

Oracle, SQL Server, IBM Db2, MySQL

Networking & telecom

Connects people and machines

Cisco, Huawei, Ericsson, Nokia, AT&T

Internet platforms

Runs the firm's websites

Web hosting, Microsoft, Oracle, Dell

Consulting & integration

Experts who make it all fit together

Accenture, IBM, Infosys, Wipro

04

Mobile, virtualization and the cloud

Computing delivered as a service

The mobile platform and BYOD

The mobile platform

7 bn+

smartphone users worldwide

4.5 h

a day U.S. adults spend on their phones

Apps lock users in: an iPhone app does not run on Android, so switching platforms gets more costly with every app you add.

BYOD — Bring Your Own Device

+ Benefits

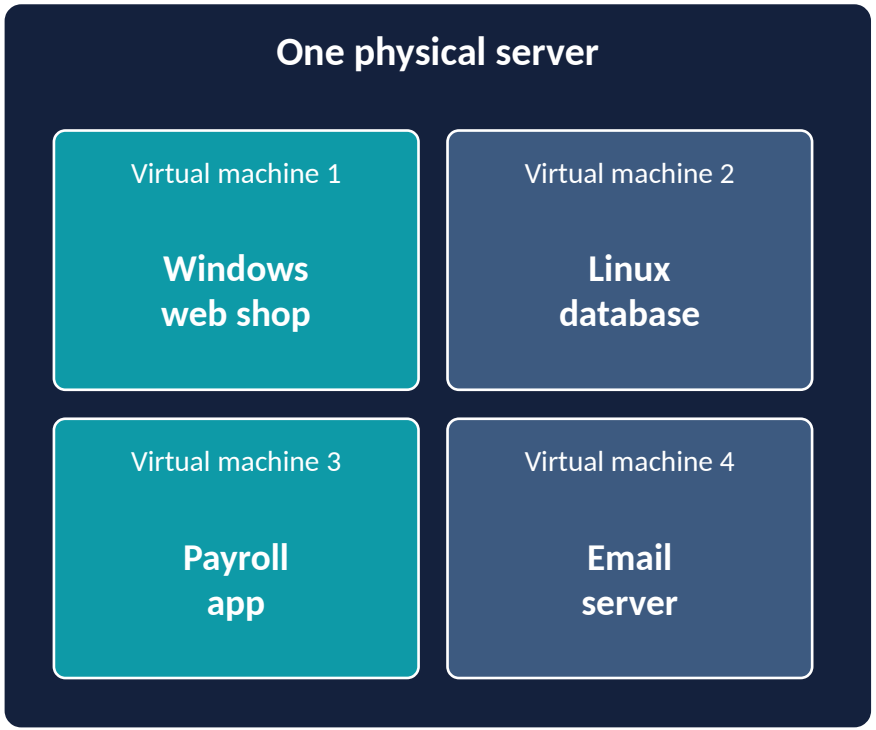
- Flexibility and convenience
- Lower hardware costs for the firm

– Risks

- Lost or stolen phones expose data
- Many device types to manage
- Distraction from social media

Fix: Mobile Device Management (MDM) — track, update and remotely wipe devices.

Virtualization: one machine, many "virtual" machines



Typical server use

15–20% → **70%+**

without → with virtualization

- Fewer physical machines to buy
- Less data center space and energy
- Easier to manage from one place
- The technology that makes the cloud possible

Analogy: one building divided into many apartments.

What is cloud computing?

Cloud computing: computing power, storage and software rented over the Internet as a shared pool of resources.



On-demand self-service

Get servers or storage yourself, instantly



Access from anywhere

Use it from any device: laptop, phone, tablet



Resource pooling

Many customers share one big pool of machines



Rapid elasticity

Scale up or down in minutes as demand changes



Measured service

Pay only for what you actually use



Analogy: like electricity — you plug in and pay for what you use instead of running your own power plant.

Three types of cloud service: IaaS, PaaS, SaaS

	Own data center	IaaS	PaaS	SaaS
Application	You	You	You	**Provider**
Platform & tools	You	You	**Provider**	**Provider**
Servers & storage	You	**Provider**	**Provider**	**Provider**
Building & power	You	**Provider**	**Provider**	**Provider**
Examples	Your own servers	Amazon EC2 and S3	Microsoft Azure, Salesforce Platform	Microsoft 365, Gmail, Salesforce CRM
Pizza analogy	*Homemade pizza*	*Take & bake*	*Pizza delivered*	*Eat at a restaurant*

Who owns the cloud? Four cloud models



Public cloud

Owned by a provider, shared by many customers

Used by: Startups, websites, Google Drive, Dropbox



Private cloud

Run for one organization only — more control

Used by: Banks, governments, strict privacy needs



Hybrid cloud

Private for core systems + public for the rest

Used by: Most large companies



Industry cloud

Cloud tailored to one sector's rules and needs

Used by: Healthcare, finance, government

Cloud computing: benefits and drawbacks

90%+

of companies worldwide use some form of cloud services (2024)

Benefits

- No big up-front investment
- Pay only for what you use
- Scale up or down quickly
- Provider does the maintenance
- Small firms get big-firm technology

Drawbacks

- Your data sits with someone else
- Security and privacy worries
- An outage can stop your business
- Bills can run out of control
- Old (legacy) systems are hard to move

Edge computing: process data close to where it is born



Why use the edge?

- Less data travels over the network
- Faster: milliseconds matter for self-driving cars, factories, trading
- Keeps working with a weak connection

Edge AI: AI on the device itself

- Face unlock on your phone
- Live translation without the Internet
- Smartwatch health alerts

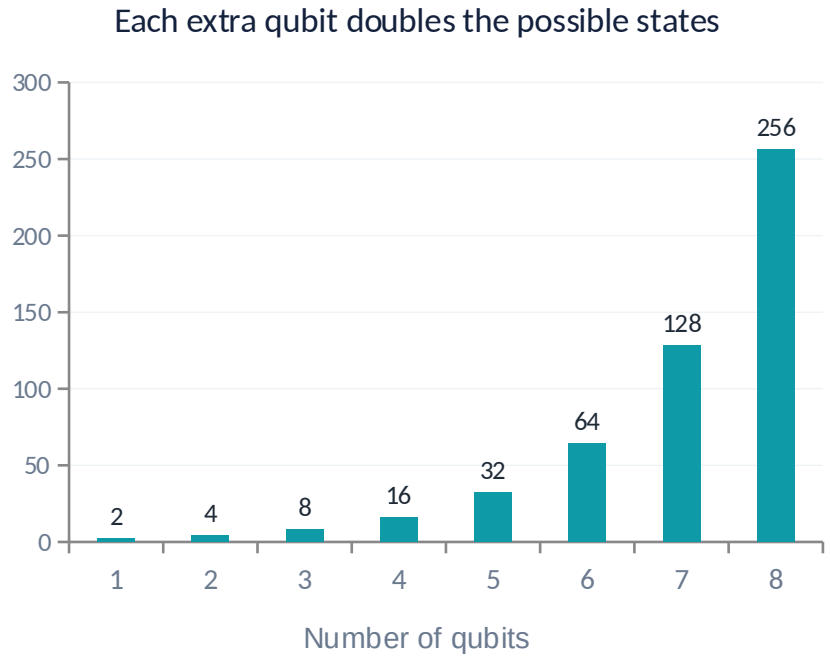
Quantum computing: the next leap?

Bit 0 or 1

Qubit 0, 1 — or both at the same time

Big challenges

- Must be cooled to −459°F, near absolute zero
- Tiny disturbances cause errors
- Could break today's encryption



Green computing: is the cloud good for the planet?

>100×

more power used by a data center than by an office building

~40%

of data center electricity goes to cooling

2–4%

of global greenhouse gas emissions come from IT — like aviation

626,000 lb

of CO_2 to train one large AI model ≈ 300 NY–SF round-trip flights

Green computing practices

- Energy-efficient chips and devices
- Renewable power: wind, hydro, solar
- Virtualization and shared cloud data centers
- Recycling e-waste (about 50 million tons a year)
- Remote work: less commuting

Software: building blocks and ways to get it

Today's software building blocks

Open source: free code anyone can improve — Linux, Firefox

HTML5 & JavaScript: interactive web pages with video, no plug-ins

Java: write once, run on almost any device

Web services (XML): different systems talk through shared standards — e.g. booking a Dollar rental car on Southwest Airlines' website

Three ways to get software



Buy a package

Ready-made, e.g. payroll software



Rent it as SaaS

Pay a subscription, use it online

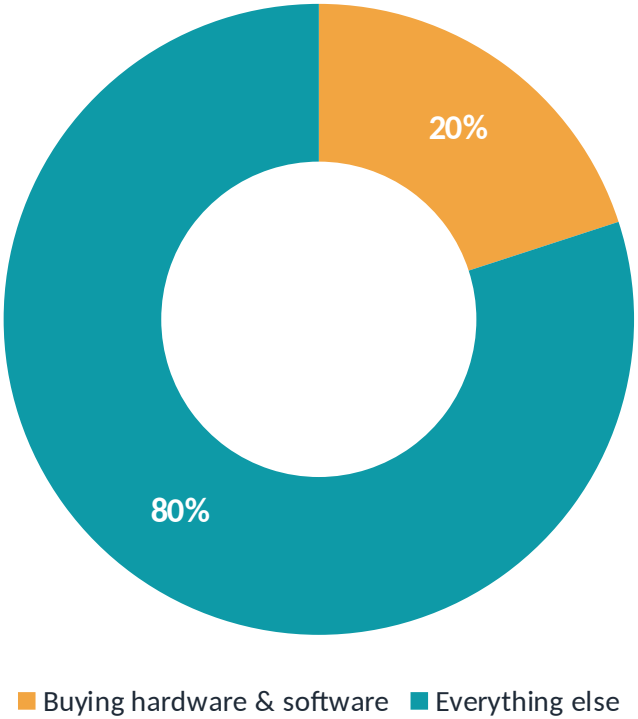


Outsource

Hire a firm, often abroad, to build it

Always sign an SLA: a contract fixing the service level you get.

Smart IT decisions: total cost of ownership (TCO)



The price tag is only the tip of the iceberg

- Installation
- Training
- Support & help desk
- Maintenance & upgrades
- Networks & equipment
- Downtime
- Space & energy

up to 3×

A desktop's total cost can reach three times its purchase price.

Key takeaways and questions for discussion

1. Hardware = CPU, memory, storage, GPU; software tells it what to do.
2. CPUs do varied tasks in sequence; GPUs do simple tasks in parallel — ideal for AI.
3. Infrastructure evolved from mainframes to cloud & mobile; all eras still coexist.
4. The cloud sells computing like a utility: IaaS, PaaS, SaaS; public, private, hybrid.
5. Good IT decisions look at total cost of ownership, not the price tag.

? Discuss

- Why is the GPU at the heart of the AI boom?
- Should a small local shop use the cloud? And a large bank?
- Is cloud computing good or bad for the environment?